Rajat Arora

Austin, TX | +1 732-874-1211 | rajat1998@gmail.com | GitHub: RajatA98 | LinkedIn | US Citizen

SUMMARY

Senior AI engineer building production RAG and agent systems that do not break quietly. Full-time architect of an enterprise RAG platform serving 29 engineering projects, with hybrid retrieval, reranking, governed ingestion, and eval-driven quality loops. Former verification engineer bringing regression testing, coverage discipline, and failure analysis into AI reliability.

CORE SKILLS

AI Systems: RAG, agents, LangGraph, ReAct, MCP, tool calling, eval harnesses, hallucination testing, regression gating

Retrieval/Data: Hybrid search, reranking, vector DBs, FAISS, Qwen embeddings, corpus filtering, Databricks, RBAC ingestion

Engineering: Python, TypeScript, SQL, C/C++, SystemVerilog, Docker, CI/CD, Azure, AWS, GCP

EXPERIENCE

AI Engineer Consultant - Lummus Technology | Full-time Contract | **Jun 2026 - Present**

- Turned a week of manual searching into a single query answered in seconds, as sole architect of a retrieval system that combines keyword and semantic search, then reranks the top matches before answering
- Cut noise from the document set by tracing 15 sources of duplication and bad text, then designing a four-stage filter projected to remove nearly 70 percent of it
- Brought three separate company systems into one searchable document set, with incremental updates so new material flows in without rebuilding from scratch
- Built the shared retrieval layer behind 2 internal AI applications used across 29 engineering projects, with role-based access controlling who sees which documents

AI Engineer - Gauntlet AI | Selected top 1% program | **Feb 2026 - Apr 2026**

- Shipped a brokerage-connected AI agent across 9 tools with financial PII guardrails and a 125-case eval harness for tool selection, hallucination, and action accuracy
- Earned top-6 cohort selection for a 40K+ viewer showcase by prototyping an sEMG wearable across ESP32 firmware, Flutter, and Python ML
- Delivered 15+ AI systems in 10 weeks across agents, RAG pipelines, MCP integrations, LangGraph workflows, eval harnesses, and observability

Design Verification Engineer II - BAE Systems | Validation and reliability infrastructure | **Nov 2022 - Nov 2025**

- Reduced manual debug time 40% by building self-checking validation systems that converted complex failures into reproducible signals with monitors, scoreboards, and reference models
- Accelerated reusable OOP testbench development with in-house AI tools, drafting and refining SystemVerilog/UVM agents, drivers, monitors, and scoreboards to reach 100% coverage across 4 protocols
- Expanded company-wide DV reuse by adding Hamming encoding to a shared RF data-bus protocol library, strengthening error detection across team testbenches

ENTREPRENEURIAL TECHNICAL BUILDS

Director of Product & Technical Lead - CloudCred | Early-stage venture, part-time | **May 2026 - Present**

- Shipped mobile, web, and backend apps in 4 weeks by architecting a monorepo with shared database schema, shared business logic, and 700+ automated tests
- Prevented silent reward-balance regressions by building a double-entry ledger in TypeScript with bigint accounting, zero-sum invariants, and test-first money movement logic
- Gated every pull request across an 8-engineer intern team by writing the TDD standard, enforcing 100% coverage, and turning review into a release-quality checkpoint
- Drove a from-scratch build to release candidate by owning product scope, system architecture, backend implementation, and cross-platform delivery decisions

Founding Engineer - Traveling Chess (Knight2) | Early-stage venture, part-time | **Dec 2025 - Present**

- Reduced tournament host workload 50% by building the event system end to end: QR match recording, self-reported results, automatic pairings, and verified Glicko-2 rating updates
- Took a large production app from stuck-in-review to live with 100+ first-month downloads by hardening DB access rules and mobile autofill

EARLIER EXPERIENCE & EDUCATION

Associate Design Verification Engineer - L&T Technology Services | Dec 2021 - Dec 2022

- Converted manual FPGA validation into release-gating regressions by automating verification workflows in C
- Increased defect detection consistency by replacing one-off inspection steps with scripted checks for FPGA behavior

Associate Software Engineer - Infosys | Nov 2020 - Dec 2021

- Protected daily production data availability by debugging SQL pipeline failure and tuning backend data workflows
- Reduced recurring production data issues by tracing failures across query logic, schedules, and data-quality breaks

Purdue University - B.S., Computer Engineering | West Lafayette, IN | **2020**